

Cultural Preservation Simulation

An Agentic Simulation of Cultural Preservation and Erosion

Track: B — Applied Agent Experience

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1. Project Summary

This project builds an interactive agentic simulation that models how cultural identity diminishes or is preserved among first-generation immigrants who migrate during adolescence or other early developmental stages. Users set a starting profile which include variables like, age at migration, household language environment, neighborhood composition, and social circle and the simulation traces how key dimensions of their cultural identity (language fluency, food and traditions, social affiliation, and self-presentation) evolve across four life stages. The system is powered by three coordinated AI agents: a Profile Agent that builds the user's cultural baseline and generates a starting score, a Drift Simulation Agent that models identity change at each life stage using environmental and relational pressures, and a Reflection Agent that synthesizes findings into a cultural preservation report. The goal is not to prescribe outcomes but to reveal which factors most accelerate or slow cultural erosion, and what interventions (familial, social, or environmental) produce meaningfully different trajectories.

2. Problem Statement

First-generation migrants who arrive during early adolescence, roughly ages before 16, have a unique developmental window. They may be old enough to carry a formed cultural identity from their country of origin, but still young enough to be fundamentally reshaped by their new social environment. Over time, key markers of cultural identity decline at different rates: native language fluency declines as the host language becomes dominant, traditional food and cultural practices fade as peer and social norms shift, and self-identification with the heritage culture weakens as immigrants adapt to survive socially. In some cases this produces not just assimilation but active distancing as they may reject or express prejudice toward newer arrivals from their own home country. The long-term consequence is generational cultural loss where there's a loss of heritage that increases with each generation, and increased geographic distance from the origin community. Despite the significance of this process, there are few accessible tools that make its dynamics visible, interactive, or researchable outside of academic ethnographic studies.

3. Target Users

Primary user: First-generation migrants, particularly those who migrated prior to adulthood, who want to understand and reflect on their own cultural trajectory or explore counterfactual scenarios.

Secondary users: Cultural and ethnographic researchers studying assimilation patterns; Educators and community organizations working with immigrant populations, interested in heritage preservation.

Overall, this is just for the broader diaspora community, for whom cultural erasure is a lived and often underdiscussed phenomenon.

4. Why is this an Agentic Problem?

This problem is well suited to an agentic approach for three reasons.

First, cultural drift is not a single linear process because it involves multiple simultaneous dimensions (language, food, social affiliation, self-presentation) that influence each other and respond differently to different pressures. A single static model cannot capture that interdependence.

Second, the simulation needs to make context-sensitive decisions at each life stage based on the user's evolving profile, not just retrieving information, but reasoning about how a specific combination of factors produces a specific trajectory.

Third, the system benefits from distinct roles: one agent that understands the user's starting conditions, one that models change over time with appropriate sensitivity to compounding effects, and one that synthesizes and interprets results. Each role requires different reasoning behavior, so a multi-agent coordination would be the most suitable architecture.

5. Initial Architecture Concept

The system follows a sequential three-agent pipeline:

User Input → Profile Agent → Drift Simulation Agent → Reflection Agent → Final Report

The state is passed between agents as an explicit JSON schema to ensure the system transfers structured, auditable data at each handoff. The user sets starting conditions before the simulation runs and receives the final report at the end. There are no mid-simulation intervention points by design because the drift is modeled as a process that unfolds through accumulated environmental and social pressure, not through discrete conscious choices. Users may re-run with modified variable settings to explore counterfactual scenarios.

6. Simulation Variables

I'm working with 7 key variables

Variable	Definition	Range
Cultural Anchor Score	Formative years spent in country of origin before migration	Low / Medium / High
Age at Migration	Age at which the individual relocated to the host country	Slider 0 to 16
Heritage Language Use	Frequency of native language use at home and socially	0–100
Social Homophily	Preference for co-ethnic vs. multicultural social integration	0–100
Neighborhood Cultural Density	Proportion of co-ethnic community in immediate environment	Low / Medium / High
Institutional Anchors	Presence of cultural churches/temples, cultural orgs in school.	Present / Absent
Intergenerational Transmission Pressure	How actively parents or elders enforce cultural practices at home	0–100
Cultural Practice Frequency	how often traditional food, celebrations, and customs are actively maintained	Low / Medium / High

7. SCOPE BOUNDARIES

In scope:

- Migration occurring during/prior to adolescence with simulation ending at approximately age 25

Stage	Approximate Age Range
Arrival	Age at migration → 1 year after
Adolescence	Through end of high school
Early Adulthood	College years / early independence(18-22)

Establishing Life Age 23–25

- Four main cultural dimensions: Language, Cultural Practice, Social Affiliation, Self-Presentation
- Seven user-configurable variables

Out of scope:

- Adult migration (age 18 and above at time of arrival)
- Macroeconomic and political push/pull factors driving migration decisions
- Return migration or transnational identity patterns
- Mental health or psychological dimensions of migration

8. Multi-Agent System Canvas

System Name: Cultural Preservation Simulation

System Purpose: To model how first-generation immigrant adolescents experience cultural preservation or erosion over time, based on a configurable set of environmental, social, and familial variables. The system produces a stage-by-stage cultural trajectory and a final preservation report that identifies which factors most influenced the outcome.

How the system works end to end: The user sets their starting profile using 7 variables. The Profile Agent interprets these inputs and constructs a structured cultural baseline. The Drift Simulation Agent then processes this baseline through four life stages — Arrival, Adolescence, Early Adulthood, and Established Life — modeling how each cultural dimension shifts at each stage given the active pressures. The Reflection Agent receives the full trajectory and produces a narrative synthesis identifying what was preserved, what was lost, what the turning points were, and what changes to the input variables would have produced different outcomes.

Agents and roles:

Agent	Role	Input	Output
Profile Agent	Builds cultural baseline from user inputs	7 user-set variables	Structured cultural profile+ Risk Factors

Drift Simulation Agent	Models cultural change across 4 life stages	Cultural profile + stage pressures	Stage-by-stage drift scores per dimension + Failure Log
Reflection Agent	Original profile+ Synthesized trajectory insights + failure log	Full drift trajectory	Narrative preservation report + Cultural Preservation Score (0–100)

Coordination logic: Sequential handoff. Profile Agent runs first and passes its output to the Drift Simulation Agent. The Drift Simulation Agent processes each life stage in order, passing its accumulated trajectory to the Reflection Agent. No agent runs in parallel. The human sets conditions at the start and receives output at the end, there is no mid-simulation intervention, but the user may re-run with adjusted variables to compare trajectories.

Where agency lives: Agency lives mainly in the Drift Simulation Agent, which must reason about how a specific interaction and combination of variables produces culturally plausible outcomes at each life stage. It is agentic because it is not retrieving pre-written outcomes but it generates context-sensitive trajectories. The Reflection Agent also exercises judgment in identifying turning points and counterfactuals rather than simply summarizing scores.

Memory and state: The cultural profile and drift scores are passed explicitly between agents as structured state. Each agent receives the full history of prior outputs so context is never lost across the pipeline.

Human in the loop: The user sets starting conditions before the simulation runs and can re-run with modified variables to explore counterfactual scenarios. There is no intervention point mid-simulation by design because the system models drift as a process that unfolds without constant conscious control to reflect the sociological reality of cultural drift.

Success looks like: A user who sets two different profiles for example, one with high Cultural Anchor Score and strong Institutional Anchors, and one with low scores on both, receives meaningfully different trajectories that reflect real-world patterns, and walks away understanding which variables drove the divergence.

9. AGENT CANVAS 1 — Profile Agent

Agent Name: Profile Agent

This agent is the starting point of the simulation. It receives the user's raw inputs of the seven variable settings they chose and translates them into a structured, culturally meaningful baseline

that the rest of the simulation can reason from. It also fills in contextual details that the user didn't explicitly set but that logically follow from their inputs. For example, if a user sets age at migration to 12 and Neighborhood Cultural Density to high, the Profile Agent infers that early peer relationships are likely to be co-ethnic, which shapes the starting social composition going into the first life stage.

What it receives:

- Age at migration
- Cultural Anchor Score (developmental years in origin country)
- Heritage Language Use at home
- Social Homophily preference
- Neighborhood Cultural Density
- Institutional Anchors (presence of church, cultural org, etc.)
- Intergenerational Transmission Pressure (parental enforcement of cultural practices)
- Cultural Practice Frequency (food, celebrations, customs)

What it produces: A structured cultural baseline containing:

- A starting score (0–100) for each of the four drift dimensions: Language, Cultural Practice, Social Affiliation, Self-Presentation
- A brief narrative description of the person's cultural starting point
- A list of the strongest protective factors and the strongest risk factors based on the input combination
- A readiness assessment — how resilient or vulnerable this profile is entering the simulation

What decisions it makes:

- How to weight combinations of variables that interact with each other (e.g. high Institutional Anchors partially offsets low Neighborhood Cultural Density)
- How to translate qualitative inputs into quantitative starting scores across four dimensions
- What contextual inferences are reasonable given the input profile

Example output (for a user who migrated at age 12 with strong family and institutional anchors but low neighborhood density):

Starting Cultural Profile: This individual arrives with a strong cultural foundation shaped by 12 formative years in their country of origin. Heritage language fluency is high (score: 85) and cultural practice frequency is strong within the home (score: 78). However, low neighborhood cultural density and moderate social homophily suggest early exposure to multicultural peer environments, creating vulnerability in social affiliation (score: 62) and self-presentation (score: 70). Key protective factor: active institutional anchor (religious community). Key risk factor: absence of co-ethnic peer community in immediate environment.

Failure modes:

- Conflicting inputs (user sets high Cultural Anchor Score but very young age at migration — the agent must flag and reconcile this)
- Over-inferring context that the user didn't intend
- Producing starting scores that are too uniform to generate interesting drift trajectories

10. AGENT CANVAS 2 — Drift Simulation Agent

Agent Name: Drift Simulation Agent

This is the agentic engine of the simulation. It takes the structured cultural baseline from the Profile Agent and models how each of the four cultural dimensions — Language, Cultural Practice, Social Affiliation, and Self-Presentation — evolves across four life stages: Arrival, Adolescence, Early Adulthood, and Established Life (ending at approximately age 25). At each stage it reasons about which variables are exerting the most pressure on each dimension and produces updated drift scores that reflect realistic, compounding cultural change.

What it receives:

- Structured cultural baseline from the Profile Agent (starting scores across four dimensions)
- The full set of user-defined variables as active pressures
- Life stage parameters (the environmental and social context typical of each stage)

What it produces:

- Updated drift scores for each cultural dimension at each of the four life stages
- A brief explanation of what drove the change at each stage
- Identification of the single most influential variable at each stage
- A complete stage-by-stage trajectory table that gets passed to the Reflection Agent

What decisions it makes:

- How much each variable influences each cultural dimension at each stage
- How compounding works — drift that happens in Adolescence shapes the starting conditions for Early Adulthood
- How protective factors slow drift and risk factors accelerate it
- When a dimension crosses a critical threshold indicating significant or irreversible erosion

Example output (Adolescence stage, same profile as before):

Stage 2 — Adolescence: Heritage language fluency declines moderately (85 → 71) as the host language becomes dominant in school and peer environments. Cultural practice frequency holds relatively stable (78 → 72) due to active parental enforcement at home. Social affiliation shifts more sharply (62 → 44) as multicultural peer integration accelerates and co-ethnic social ties weaken. Self-presentation begins to show early assimilation markers (70 → 58) including informal name adaptation. Primary driver of change at this stage: low neighborhood cultural density combined with high multicultural social exposure.

Failure modes:

- Producing trajectories that are too uniform across different input profiles — if two very different profiles produce similar outcomes, the simulation is not reasoning meaningfully
- Over-weighting a single variable and producing extreme outcomes
- Failing to model compounding correctly — treating each stage as independent rather than cumulative
- Generating culturally inaccurate patterns for non-Nigerian origin profiles

11. AGENT CANVAS 3 — Reflection Agent

Agent Name: Reflection Agent

This agent is the analyst and interpreter. It receives the complete drift trajectory from the Drift Simulation Agent and produces a final cultural preservation report. Rather than just summarizing what happened, it identifies why it happened, what the turning points were, which variables had the most influence, and what would have produced a different outcome. This is the output the user actually reads and learns from.

What it receives:

- The full stage-by-stage drift trajectory across all four cultural dimensions
- The original user profile and variable settings
- The Profile Agent's initial protective and risk factor assessment

What it produces:

- A narrative summary of the full cultural trajectory
- A ranked list of the variables that most influenced the final outcome
- Identification of the critical turning point stage — when drift accelerated most
- A counterfactual analysis: what changes to one or two variables would have meaningfully altered the outcome
- A final Cultural Preservation Score (0–100) representing overall heritage retention at age 25

- A one-paragraph reflection connecting the simulation output to broader patterns of generational cultural loss

What decisions it makes:

- How to interpret the relative significance of drift across dimensions — losing language fluency vs. losing cultural practice frequency are not equally weighted losses
- How to frame counterfactuals responsibly without implying that assimilation is inherently negative
- How to balance analytical rigor with narrative accessibility for non-researcher users

Example output:

Final Cultural Preservation Report: This profile ends with an overall Cultural Preservation Score of 61/100, indicating moderate heritage retention. Language fluency experienced the steepest decline, dropping from 85 to 53 across the four stages, primarily driven by the absence of co-ethnic peer community during adolescence. Cultural practices showed the most resilience, supported by consistent intergenerational transmission pressure at home. The critical turning point was the Adolescence stage, where social affiliation shifted sharply as multicultural peer integration accelerated. Counterfactual: had Neighborhood Cultural Density been set to high rather than low, modeling suggests Social Affiliation scores would have remained 15–20 points higher across all subsequent stages, with downstream protective effects on Language and Self-Presentation. This trajectory reflects a pattern common among adolescent immigrants in culturally sparse environments which is a strong home preservation paired with accelerating external assimilation.

Failure modes:

- Producing generic reflections that do not meaningfully connect to the specific input profile
- Framing cultural drift as inherently negative rather than as a neutral dynamic with varying outcomes
- Counterfactuals that are implausible or too speculative to be useful
- Failing to distinguish between dimensions that declined gradually versus those that shifted suddenly

12. INITIAL SUCCESS CRITERIA

I'll need to do more research on possible baselines or criteria for evaluation but I've listed some tests I'll check for below:

1. Two meaningfully different input profiles produce measurably different drift trajectories across at least three of the four cultural dimensions
2. The Drift Simulation Agent correctly compounds change across stages and the outcomes at stage 4 reflect accumulated history from stages 1 through 3
3. The Reflection Agent identifies the correct primary driver of drift in at least 3 out of 5 test scenarios
4. Any type of user can read the final report and accurately identify which two variables most influenced their outcome without being told explicitly

13. Evaluation Criteria

In order to test if the agents are all working correctly, I'll be running some some specific evaluation//validation checks:

I. Sociological Divergence

- **Definition:** Test if different "Ages at Migration" settings produce different results?
- **Success Measure:** For example, If the user sets "Age: 16," the Language score should typically remain > 60%. If they set "Age: 5," it should drop below 20%.
- **Audit Method:** Run various simulations with varied ages and graph the correlation.

II. State Persistence

- **Definition:** Does the Reflection Agent actually remember what the Profile Agent said?
- **Success Measure:** The final report must correctly identify at least 3 initial variables as the "primary drivers" of the outcome.
- **Audit Method:** Compare the input JSON from Stage 1 to the final text output in Stage 3.

IV. Counterfactual Accuracy

- **Definition:** Does the reported logic make sense?
- **Success Measure:** If the Reflection Agent suggests that "Higher Institutional Support" would have saved the language, a re-run with that variable turned up should actually show that improvement.

V. Narrative Coherence

- **Definition:** Is the Reflection Report professional and insightful?
- **Success Measure:** A user (even non-researchers) should be able to read the report and identify their own cultural risk factors without looking at the raw data.

14. INITIAL RISKS AND CONCERNS

1. **Variable Interaction Complexity:** LLMs may struggle with 7 interacting variables and some variables interact in non-obvious ways that are hard to model through prompting alone.

Mitigation: Use a "Chain of Thought" reasoning step and explicit guidelines for common interaction patterns for the Drift Agent at each stage.

2. **Normative framing** — there is a risk that the simulation implicitly frames cultural assimilation as failure and preservation as success. The Reflection Agent must be carefully prompted to treat both as neutral outcomes with tradeoffs.

Mitigation: explicit prompt design and evaluation criterion for tone.

3. **Generic outputs** — the biggest usability risk is that two very different profiles produce similar-sounding narratives.

Mitigation: evaluation will specifically test profile pairs designed to be maximally different and score output distinctiveness.

15. TEAM PLAN

It's an individual submission, so all components are led by me to the best of my ability.